Feature Engineering and Model Evaluation for E-commerce Dataset

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Abstract

This report presents a detailed study on feature engineering and model evaluation for a multi-class classification problem using an e-commerce dataset. Three classifiers – Support Vector Machine (SVM), Logistic Regression, and Decision Tree – were evaluated with three different feature engineering techniques. Performance metrics such as accuracy, macro-average F1 score, training, and inference times were compared. Additionally, hyperparameter tuning and OneVsRest evaluation were conducted to identify optimal model configurations and to assess model robustness across different classes.

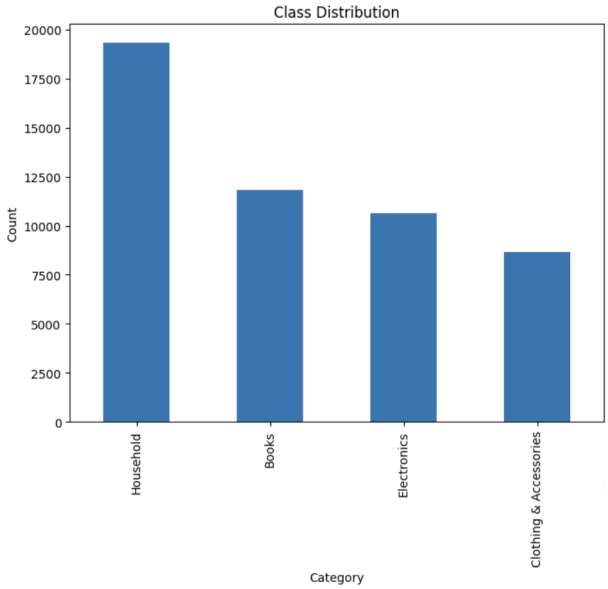

Figure 1: Class Distribution

1 Introduction

In this assignment, we aim to solve a 4-class classification (Household, Books, Electronics, Clothing& Accessories) problem using an e-commerce dataset that includes item categories and descriptions. Class Distribution like the Figure 1. We employed three classifiers (SVM, Logistic Regression, Decision Tree) and evaluated them with three different feature engineering techniques: Bag-of-Words (BoW), TF-IDF, and N-gram. The dataset was split into 70% training, 10% validation, and 20% testing, ensuring reproducibility by using a fixed random seed.

The training, validation, and test set sizes are as follows:

- Training set size: 35296
- Validation set size: 5042
- Test set size: 10086

2 Feature Engineering Techniques

Three feature engineering methods were applied to the text data:

Feature Engineering

For the classification task, three distinct features were developed using different vectorization techniques:

- ***CountVectorizer* for Bag-of-Words**: from scikit-learn. By transforming the text input into a matrix of token counts, this method enables the model to record information on word frequency. In the *review/text* column, we eliminated rows with missing data then used the *CountVectorizer* to choose the 3,000 most common features.
- **TF-IDF using TfidfVectorizer**: The second feature employed the *TfidfVectorizer*,which determines the Term Frequency-Inverse Document Frequency (TF-IDF) values for every

word in the review text. This is done in natural language processing and information retrieval tasks to evaluate a word's significance inside a document and its prevalence throughout the corpus. In *TfidfVectorizer*, Higher priority is given to uncommon words in the entire corpus that often occur in a particular document. We chose 3,000 max features that were kept for this project in this part.

- **N-gram**: A feature that considers the sequence of words to capture contextual information. Using `CountVectorizer` with `ngram_range=(1, 2)`, it generates a word frequency vector containing both 1-gram and 2-gram features. Specifically:

  1-gram: 1-gram represents the frequency of individual words. For example, in this description, words like "Paper", "Design", "Framed", "Wall" will be counted separately.

  2-gram: 2-gram represents the combination of two adjacent words. For instance, "Paper Plane", "Plane Design", "Design Framed", "Framed Wall" are combinations of two adjacent words that will also be counted.

  `CountVectorizer` will generate a bag-of-words model that contains these 1-gram and 2-gram terms along with their respective frequencies. For example, if "Paper Plane" appears once and "Design Framed" appears once, these 2-grams will be recorded and become part of the feature vector.

  Example of Generated N-grams:

  Original Text:

  > Paper Plane Design Framed Wall Hanging Motivational Office Decor Art Prints (8.7 X 8.7 inch) - Set of 4 Painting made up in synthetic frame with uv textured print which gives multi effects and attracts towards it. This is a special series of paintings which makes your wall very beautiful and gives a royal touch. .... It would be treasured forever by whoever your lucky recipient is. Liven up your place with these intriguing paintings that are high definition hd graphic digital prints for home, office, or any room.

  Based on this text content, here are examples of 1-gram and 2-gram features:

1-gram (Single Word Features)	**2-gram (Word Pair Features)**
"Paper"	"Paper Plane"
"Plane"	"Plane Design"
"Design"	"Design Framed"
"Framed"	"Framed Wall"
"Wall"	"Wall Hanging"
"Hanging"	"Motivational Office"
"Motivational"	"Office Decor"
"Office"	"Decor Art"
"Decor"	"Art Prints"
"Art"	"Printed Painting"
"Prints"	"high definition"
"Painting"	"unique painting"
"technology"	"true colours"
"colors"	"to hang"
"beautiful"	"special series"
"unique"	"to your"
"recipient"	"etc."
"etc."	

3 Results and Analysis

3.1 Performance Comparison

The accuracy, macro-average F1 scores, training time, and inference time for each model with different features are shown below.

Model	Accuracy	Macro F1	Training(s)	Inference(s)
LR (BoW)	0.9740	0.9742	17.22	0.0125
LR (TF-IDF)	0.9676	0.9678	17.55	0.0110
LR (N-gram)	0.9750	0.9752	138.17	0.0571
DT(BoW)	0.9545	0.9553	34.85	0.0163
DT(TF-IDF)	0.9477	0.9482	42.84	0.0423
DT(N-gram)	0.9535	0.9544	152.72	0.0699
SVM(BoW)	0.9734	0.9735	13.63	0.0145
SVM(TF-IDF)	0.9783	0.9786	2.22	0.0145
SVM(N-gram)	0.9751	0.9750	36.68	0.0434

Table 1: Performance Comparison of Different Models and Feature Engineering Techniques

Best Model Selection and Optuna Hyperparameter Optimization (Commented Out)

Although the hyperparameters were predefined, Optuna was originally used to optimize these hyperparameters. Optuna systematically explored a range of hyperparameters and iteratively tested different combinations to maximize the validation accuracy, ultimately identifying the best parameter values.

The best-performing models were further optimized using hyperparameter tuning:

- **Logistic Regression**: $C = 53.998$, Solver: lbfgs
- **Decision Tree**: max_depth = 32, min_samples_split = 6
- **LinearSVC**: $C = 6.833$, max_iter = 7783

Model Results Summary

Results Explanation

Logistic Regression

- **Best Performance**: Using TF-IDF features achieved the best accuracy of 0.9804 and a macro F1 score of 0.9807, with relatively low training and inference times.
- **High Training Time for N-gram**: The model with N-gram features required the longest training time (555 seconds) but still achieved high accuracy and F1 scores.

Decision Tree

- **Consistent Performance Across Features**: All three feature sets had similar performance, with accuracy ranging from 0.8919 to 0.8924.
- **Fastest Training with BoW**: Using BoW features had the shortest training time (13.8 seconds), while N-gram features required a longer training time (37.1 seconds).

LinearSVC

- **Best Model with TF-IDF**: The TF-IDF model had the shortest training time (5.5 seconds) and achieved the best accuracy of 0.9799 and macro F1 score of 0.9802.
- **Increased Training Time with N-gram**: The N-gram model significantly increased training time (218.7 seconds) but achieved slightly lower accuracy than the TF-IDF model.

Overall Analysis

- **Optimal Model and Feature Combination**: In terms of accuracy and macro F1 score, **Logistic Regression (TF-IDF)** and **LinearSVC (TF-IDF)** are the best choices. Both have high accuracy and low training and inference times.
- **Recommended Model for Quick Training**: For tasks requiring fast training, **LinearSVC (TF-IDF)** is the best option, with the shortest training time and excellent performance. If longer training time is acceptable, **Logistic Regression (TF-IDF)** provides similarly high accuracy and stable performance.

3.2 OneVsRest Evaluation

To assess the robustness of the model across different classes, we used the OneVsRest approach. The average accuracy and macro-average F1 scores for each class are as follows:

- **Class 1**: Accuracy = 0.9781, Macro F1 Score = 0.9689
- **Class 2**: Accuracy = 0.9869, Macro F1 Score = 0.9768
- **Class 3**: Accuracy = 0.9759, Macro F1 Score = 0.9628
- **Class 4**: Accuracy = 0.9739, Macro F1 Score = 0.9724

Average Accuracy (OneVsRest): 0.9787
Average Macro F1 Score (OneVsRest): 0.9702

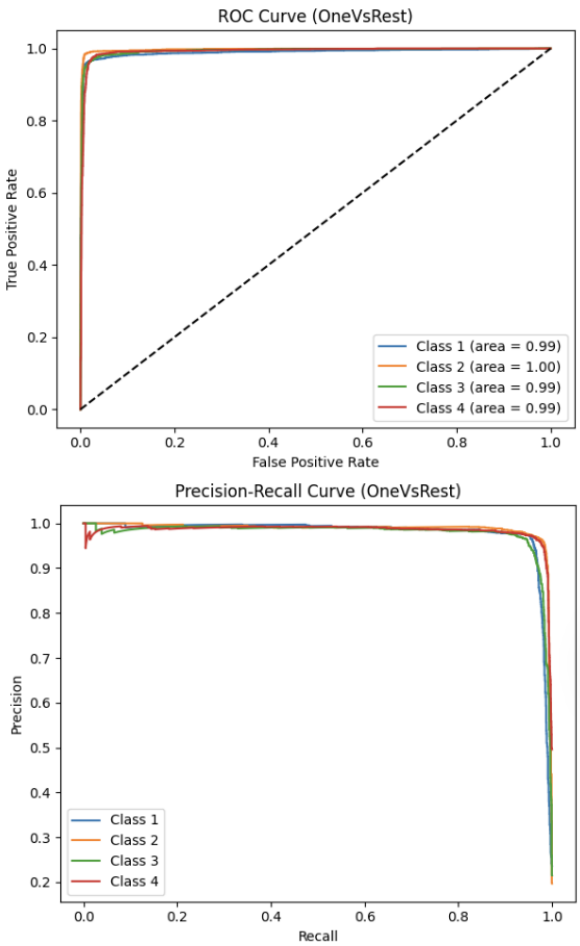

Figure 2: ROC and Precision-Recall Curves (OneVs-Rest)

4 Conclusion

In this study, we evaluated multiple feature engineering techniques and classifiers for a multi-class classification problem. SVM with TF-IDF achieved the highest accuracy and macro F1 score, making it the recommended model for this task. Logistic Regression with N-gram features also performed well, albeit with a longer training time. The OneVsRest evaluation provided further insights into the model's performance across different classes, confirming the robustness of the SVM with TF-IDF feature engineering.